22·05·2025(E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: TYBTech - COMP	Class: SY	Semester: V (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMPUTER	
Course Code: 116U01C501	Name of the Course: Software Engineering	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Define EVA and explain how it helps in measuring project performance.	5
ii)	List and describe the five maturity levels defined in the Capability Maturity Model (CMM).	5
iii)	How is the development phases handled differently in RAD and Waterfall models?	5
iv)	Using Function Point Analysis, calculate the total Function Point (FP) value based on the following information. Assume all components are of average complexity (given in bracket). Given: • Number of User Inputs = 50 (4) • Number of User Outputs = 40 (5) • Number of User Enquiries = 35 (4) • Number of User Files = 6 (10) • Number of External Interfaces = 4 (7) Calculate the total Function Point (FP) value. Show all calculation steps.	5
v)	Explain the role of product metrics in project planning and estimation.	5
vi)	State the different phases of the Software Development Life Cycle (SDLC).	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain functional and non-functional requirements in software engineering. Give two examples of each.	5

ii)	Explain various techniques used for requirement elicitation in software engineering.	5
	OR	
Q2 A	Draw a use case diagram for Engineering college portal kindly state the attributes, requirements and constraints clearly.	10
Q 2 B	Solve any **One**	**10**
i)	For most of the bank's ATMs, the customer is authenticated by inserting a plastic ATM card and entering a personal identification number (PIN). The person can use a face recogniser for authentication alternatively. Customer Authentication use case is required for every ATM transaction such as balance check and withdrawal of money. Customers may need some help from the ATM. The transactions may permit to take printout of the current balance or printing slip after the withdrawal mentioning the balance amount. Draw ANY TWO TYPES of diagram: a. USE CASE diagram b. Activity Diagram c. Sequence Diagram	10
ii)	What is Software Requirement Specification (SRS)? Describe the components of a good SRS document.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain the concept of software reuse. What are its advantages and challenges in modern software development? Illustrate with suitable examples.	10
ii)	Describe user interface design rules in detail.	10
iii)	Explain the concepts of cohesion and coupling in object-oriented design. Why is high cohesion and low coupling desirable? Illustrate with examples or diagrams.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Describe elements of Component Diagram with necessary diagram.	10
ii)	Classify various software risks and explain how each category can affect project success.	10
iii)	Discuss the significance of software maintenance in the software lifecycle. Describe in detail the various types of maintenance and their impact on software quality and performance.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four** (Short notes / Short question type)	**20**
i)	Principles of Testing	5
ii)	Reverse Engineering	5
iii)	Risk Identification	5
iv)	Equivalence Partitioning	5
v)	Formal Technical Review	5
vi)	Acceptance Testing	5